

Econ 131
Spring 2026

Midterm 2 Draft

Monday, April 6, 2026

Student Name:

Student ID:

Exam Instructions

- **This exam has three questions. Question 1 and Question 3 are worth 60 points each, and Question 2 is worth 30 points (150 points total). The total exam time is 75 minutes.**
- **Write answers using pens (we recommend black or blue ink), and write only in the space provided for each answer so your solutions scan properly.**
- **Do not write your solutions on pages that say “Do not write on this page”. Answers written on these pages will not be graded.**
- **Show your work.** Credit will only be awarded on the basis of what is written on the exam.
- **Affirm the academic honesty pledge below.** For those writing on a non-printed copy, please write “Academic Honesty Pledge as on exam” and sign your name. **If you do not affirm this pledge, your exam will be marked invalid.**

0. ACADEMIC HONESTY PLEDGE

I confirm that I have abided by all academic honesty rules for UC Berkeley and Economics 131. I confirm that I did not see this exam before my official exam start time. I confirm that I have not shared and will not share this exam with anyone else.

Signature: _____

1. True/False/Uncertain Questions (60 points total)

You can start your answer “True”, “False”, “Uncertain”, “Partially true”, or “Partially false”, etc., but your choice of the label does not matter much. What matters is your explanation.

- (a) Suppose that a tax reform cuts the corporate tax rate and makes investment immediately deductible, rather than deductible over time. The tax reform will increase investment.

- (b) If higher-skill individuals systematically earn higher returns on wealth than lower-skill individuals, the baseline Atkinson-Stiglitz case for a zero capital income tax becomes weaker.

- (c) Suppose a town's new data center uses a lot of electricity, which makes electricity more expensive for the town's families. This is an example of a market failure.

- (d) When pollution abatement costs are uncertain and when the marginal damage from pollution does not depend on the pollution level, tradable permits can be preferable to a corrective tax.

- (e) For private goods, social efficiency requires summing marginal rates of substitution across consumers in the same way as for public goods.

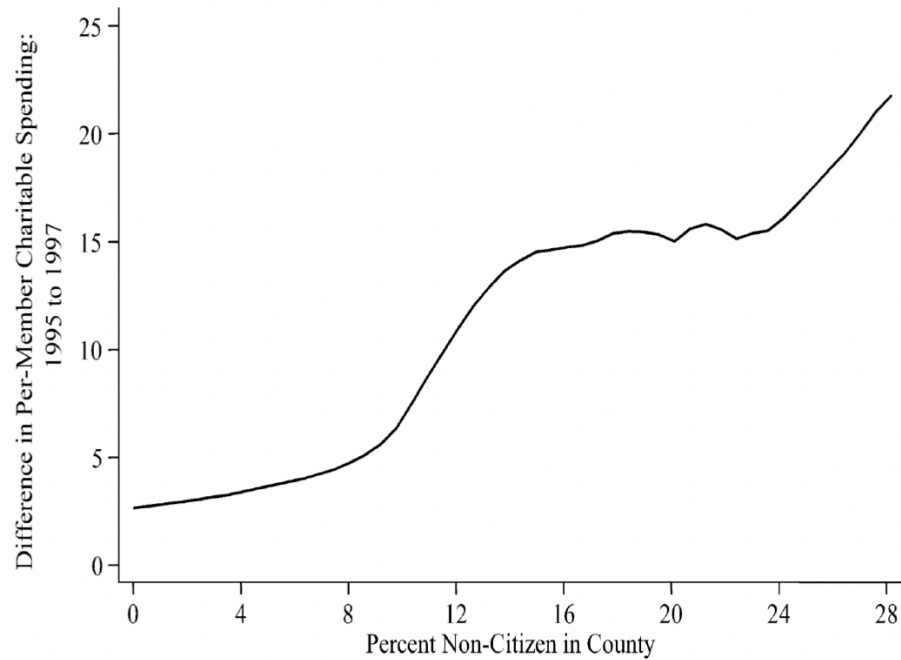
- (f) Warm-glow motives can imply less than one-for-one crowd-out of private giving by government provision.

- (g) Suppose Berkeley residents use Oakland's public playgrounds. The Tiebout model says that Oakland and Berkeley public playgrounds will nevertheless be efficiently provided since people "vote with their feet".

(h) People know the quality of local public goods.

2. Empirical Application: Hungerman (2005) and Crowd-Out of Church Charity (30 points total)

In 1996, welfare reform reduced or eliminated some transfers for non-citizens. Hungerman (2005) studies whether church-provided charitable spending rose more in places more exposed to this reform. The figure below plots the change in per-member charitable spending from 1995 to 1997 against the percent non-citizen in the county.



Source: Hungerman 2005

- (a) Explain in words the difference-in-differences style conclusion this figure is trying to suggest. What broad pattern in the figure would be consistent with the idea that counties more exposed to the welfare cuts experienced larger increases in church charitable spending? *(7.5 points)*

- (b) Based on this figure alone, do you find the causal interpretation convincing? Answer yes or no and explain briefly. What additional evidence or alternative figure would you want to see to make the claim more convincing? *(7.5 points)*

- (c) Let $Treat_c = 1$ denote the counties with a high non-citizen share of the population, and let $Treat_c = 0$ denote otherwise. Let $Post_t = 1$ denote years after the reform and $Post_t = 0$ denote years before the reform. Let Y_{ct} denote church charitable spending in county c and year t . Suppose you estimate the regression

$$Y_{ct} = \alpha + \beta Treat_c + \gamma Post_t + \delta(Treat_c \times Post_t) + \varepsilon_{ct}.$$

Using the coefficients $\alpha, \beta, \gamma, \delta$ from this regression, fill in the four cells in the following table. Each cell equals a mean. For example, the control/pre cell equals the mean value of Y_{ct} in the control counties before the reform. (7.5 points)

	Pre ($Post_t = 0$)	Post ($Post_t = 1$)
Control ($Treat_c = 0$)		
Treatment ($Treat_c = 1$)		

- (d) Which regression coefficient or coefficients equal the difference-in-differences estimate? Briefly interpret what a positive value of that coefficient or coefficients would mean in this context. (7.5 points)

3. Negative Income Tax vs. Earned Income Tax Credit (60 points total)

A city has 1,000 working-age residents. Each resident decides whether to work or not. A worker earns pre-tax labor income $z > 0$; a non-worker earns $z = 0$. Initially there are **no taxes or transfers**.

(a) The city introduces a **Negative Income Tax (NIT)**:

- Every resident receives a **guaranteed income of \$20** (even with zero earnings).
- The guarantee is **phased out at 50 cents per dollar** of earnings.
- There are **no further taxes** above the point where the guarantee is fully phased out.

At what level of earnings z does the guarantee fully phase out? Write after-tax-and-transfer income $c(z) = z - T(z)$ for each segment, and draw the budget constraint with pre-tax earnings z on the x-axis and c on the y-axis. Label the y-intercept, any kink points (with coordinates), and the slope of each segment. *(12 points)*

(b) Using the NIT from part (a), compute the **participation tax rate**

$$\tau_p = \frac{T(z) - T(0)}{z}$$

for a worker earning $z = 20$ and for a worker earning $z = 80$. Which faces the higher participation tax rate? In one sentence, explain intuitively why a system with a phased-out guaranteed income creates especially high participation tax rates for low earners. *(12 points)*

(c) Before the NIT was introduced, there were 700 workers and 300 non-workers. After the NIT is introduced:

- 100 workers who had been earning $z = 20$ each **stop working**, discouraged by the high participation tax rate.
- 500 workers who earn $z = 80$ each continue working.
- The remaining 100 workers who earn $z = 20$ each also continue working.
- The 300 original non-workers remain out of work.

Compute the government's **total cost** of the NIT (total grants paid out, net of any phase-out clawback). *(12 points)*

(d) The city replaces the NIT with an **Earned Income Tax Credit (EITC)**:

- **No payment to non-workers:** $c(0) = 0$.
- **Phase-in** ($z = 0$ to \$20): workers receive a subsidy of \$0.50 per dollar earned.
- **Plateau** ($z = \$20$ to \$40): workers receive a flat credit of \$10.
- **Phase-out** (z above \$40): the credit is reduced by \$0.25 per dollar of earnings above \$40.
- No other taxes.

At what earnings level is the credit fully phased out? Draw the budget constraint (same axes as part (a)). Label the y-intercept, all kink points (with coordinates), and the slope of each segment. (12 points)

- (e) Motivated by the EITC structure in part (d), the government considers a modest reform to the NIT from part (c): it **reduces the phase-out rate from 50% to 25%** on the first \$20 of earnings. Above \$20, the phase-out rate remains 50%. All else stays the same (the guaranteed income is still \$20). *(12 points total)*
- (i) What is the participation tax rate at $z = 20$ under this reformed NIT? Compare it to the original NIT participation tax rate from part (b). *(4 points)*

- (ii) The reform has a **mechanical cost**: the 100 workers at $z = 20$ who were already working now receive a larger net transfer. How much extra does each one cost the government? What is the total mechanical cost? (*4 points*)

- (iii) Suppose that among the 200 residents with potential earnings $z = 20$, the **extensive-margin elasticity of the number who work** with respect to the net-of-tax rate $1 - \tau_p$ is

$$\varepsilon = \frac{\% \Delta \text{workers}}{\% \Delta (1 - \tau_p)} = 2.$$

Using the original NIT in part (c) as the baseline, compute how many of the 100 workers who had dropped out would re-enter when the participation tax rate falls from 50% to 25%. Then compute the **fiscal saving** from each returning worker: compare what they cost as a non-worker versus what they cost as a worker under the reformed NIT. What is the total extensive-margin saving? Does it cover the mechanical cost from part (ii)? Explain in 1–2 sentences why this type of reform can be a “free lunch.” (4 points)

Reminder: please sign the Academic Honesty Pledge on the first page!